# Optimal Deterministic Incremental–Decremental Submodular Maximization

ZIJIAN ZENG, UCSI University, Malaysia

Incremental–decremental maximization asks for one ordering of a finite ground set such that every prefix has high utility. For two nonnegative monotone submodular functions $h$ and $g$, the utility of a prefix $S$ is $h(S) + g(E \setminus S)$, and its benchmark is the best set of the same size. Disser, Klimm, Lutz and Strubberg established a deterministic competitive upper bound of $1 + e/(e-1)$ and a structural lower bound of 2.

We give a deterministic 2-competitive algorithm for the full class, including arbitrary unequal endpoint totals. It uses $O(|E|^2)$ exact value-oracle queries and polynomial additional computation. If $h(E) \geq g(E)$, the algorithm greedily maximizes the actual objective marginal at each step, continuing through negative gains. Otherwise it swaps the functions and reverses the resulting order. The proof retains the initial objective as a budget in a submodular union comparison; endpoint orientation controls the negative-gain suffix. Combined with the existing lower bound, which we reproduce with attribution, this determines the optimal uniform deterministic constant as 2, both with unrestricted computation and in the polynomial exact-value-oracle model.

Author's Contact Information: Zijian Zeng, UCSI University, Malaysia, 1002266693@ucsiuniversity.edu.my.

## 1 The problem and the sharp guarantee

An infrastructure renewal order must maintain useful service throughout a transition: some elements already operate under a new system, while the others remain in the old system. The incremental–decremental model of Disser, Klimm, Lutz and Strubberg [DKLS26] captures this requirement by evaluating both parts of a partition. The mathematical difficulty is to choose a *single* order that performs well at every cardinality, rather than an unrelated approximate solution for each cardinality.

Let $E$ be a finite ground set of size $n$. A set function $q : 2^E \to \mathbb{R}_{\geq 0}$ is normalized if $q(\varnothing) = 0$, monotone if $q(A) \leq q(B)$ whenever $A \subseteq B$, and submodular if

$$q(A) + q(B) \geq q(A \cup B) + q(A \cap B) \qquad (A, B \subseteq E).$$

Initially assume that $h$ and $g$ have all three properties. Define

$$F(S) = h(S) + g(E \setminus S), \qquad \mathrm{OPT}_k = \max_{\substack{O \subseteq E \\ |O| = k}} F(O). \tag{1}$$

For a permutation $\pi = (e_1, \ldots, e_n)$, write $S_k = \{e_1, \ldots, e_k\}$ and $S_0 = \varnothing$. The permutation is $r$-competitive if

$$rF(S_k) \geq \mathrm{OPT}_k \qquad (0 \leq k \leq n). \tag{2}$$

This inequality convention includes zero objectives without dividing by zero. The endpoint cardinalities each have a unique feasible set.

The previous full-class bounds are 2 and $1 + e/(e-1) \approx 2.582$: Corollary 2 and Proposition 11 of [DKLS26] give the upper and lower bounds, respectively, and its Section 5 asks to close the gap. Its Proposition 17 already gives a factor of 2 when $h(E) = g(E)$. We address the entire pair class, with no equality requirement on these totals.

Theorem 1.1. *For every finite $E$ and normalized nonnegative monotone submodular $h, g : 2^E \to \mathbb{R}_{\geq 0}$, a deterministic algorithm returns one permutation satisfying*

$$2F(S_k) \geq \mathrm{OPT}_k \qquad \textit{simultaneously for every } k = 0, \ldots, n.$$

*It uses $O(n^2)$ exact value-oracle queries and polynomial additional computation in a unit-cost exact-arithmetic and comparison model. The optimal uniform deterministic competitive constant is 2 for both unrestricted algorithms and polynomial-oracle algorithms. The same constants hold for nonnegative monotone submodular inputs without normalization.*

The algorithmic upper bound is the contribution here. The impossibility of any uniform constant below 2 is prior work [DKLS26, Proposition 11]; Section 4 reproduces a coverage-family proof for completeness. Thus the result determines the exact worst-case cost of maintaining a single nested sequence for this model, with an efficient construction.

The key distinction from ordinary monotone submodular greedy analysis is that $F$ may decrease. We preserve its positive initial value in the proof rather than subtracting it. Before greedy gains become negative, that initial value pays for the loss in comparing an optimal set with its union with a prefix. Afterward, orienting the order toward the larger endpoint supplies the guarantee directly. Both parts use elementary diminishing returns; no claim of a more general nonmonotone greedy theorem is needed.

*Relation to nearby results.* The double-greedy algorithm in [DKLS26] compares separate $h$ and $g$ marginals when extending opposite ends of an order. Its exact worst-case ratio is $1 + e/(e-1)$, even with modular $g$ [DKLS26, Proposition 19 and Corollary 20]. Our algorithm instead commits to one direction and maximizes the marginal of the *combined* objective. Consequently this is a different algorithm, rather than a sharper analysis of that double-greedy rule.

For monotone incremental maximization, the subcase $g = 0$, [BBB+26] reports an upper bound of approximately 1.373 and a lower bound of 1.25. Its improved upper bound sacrifices computational efficiency and does not supply an upper bound for arbitrary pairs $h, g$. The factor 2 proved here is sharp over the pair class; it is not asserted to be sharp for that subcase. For another nearby comparison, Chen and Kuhnle [CK26] analyze curvature-dependent greedy with pruning under a fixed at-most-$k$ constraint. Pruning can remove selected elements, so that result does not itself provide one permutation at every exact cardinality. These comparisons concern the stated models and theorems, rather than all possible consequences of those papers. A bounded primary-literature check found no equivalent full-class guarantee as of November 1, 2026; this does not establish universal priority.

## 2 The algorithm

Fix a total order on $E$ for deterministic tie breaking. For a function $W$, write $W(e \mid P) = W(P \cup \{e\}) - W(P)$ when $e \notin P$. Let $H = h(E)$ and $G = g(E)$.

(1) If $H \geq G$, start with $P_0 = \emptyset$. For $i = 1, \ldots, n$ choose

$$e_i \in \arg\max_{e \in E \setminus P_{i-1}} F(e \mid P_{i-1}), \qquad P_i = P_{i-1} \cup \{e_i\},$$

breaking ties by the fixed order. Return $(e_1, \ldots, e_n)$.

(2) If $H < G$, apply the preceding greedy construction to the swapped objective $F'(A) = g(A) + h(E \setminus A)$, and return its order in reverse.

The construction always continues until every element has been selected, including when the maximum available marginal is negative. It does not query $\mathrm{OPT}_k$. For $E = \emptyset$ it returns the empty permutation. The endpoint comparison is part of the algorithm, and will be used in the negative-gain case of the proof.

There are $n(n+1)/2$ candidate evaluations. Each combined objective value uses two summand-oracle queries, and caching the current objective permits each marginal comparison with constant additional scalar arithmetic. Together with the endpoint queries this is at most $n(n+1) + 2$ summand queries for normalized inputs. Set construction and bookkeeping require only polynomial additional computation. The model permits exact arithmetic and comparisons on oracle answers at unit cost; no bit-complexity guarantee for arbitrarily encoded real answers is asserted.

## 3 Proof of the simultaneous upper bound

We first consider the orientation $H \geq G$. Complementation preserves submodularity: apply the defining inequality for $g$ to $E \setminus A$ and $E \setminus B$, and interchange union and intersection. Thus $S \mapsto g(E \setminus S)$, and hence $F$, is submodular. Although $F$ need not be monotone, its decomposition yields two useful bounds.

Lemma 3.1. *For every $O, P \subseteq E$,*

$$F(O) \leq H + G, \tag{3}$$

$$F(O) \leq F(O \cup P) + G. \tag{4}$$

Proof. Monotonicity bounds $h(O)$ by $H$ and $g(E \setminus O)$ by $G$, proving (3). Moreover,

$$\begin{aligned} F(O) - F(O \cup P) = {} & h(O) - h(O \cup P) \\ & + g(E \setminus O) - g(E \setminus (O \cup P)) \leq G. \end{aligned}$$

The first difference is nonpositive by monotonicity of $h$; the second is at most $G$ by monotonicity and nonnegativity of $g$. □

Let $a_i = F(P_i) - F(P_{i-1})$ be the gains of the greedy order. These gains are nonincreasing, including after they become negative. Indeed, for $e \notin P_i$, diminishing returns and the preceding maximal choice give

$$F(e \mid P_i) \le F(e \mid P_{i-1}) \le a_i.$$

Maximizing the left side over the remaining elements proves $a_{i+1} \le a_i$ for $i < n$. Also $F(\varnothing) = G$ and $F(E) = H$.

Fix any internal cardinality $1 \le k < n$, put $P = P_k$ and $d = a_{k+1}$, and let $O$ be any set of size $k$.

*Case 1:* $d \ge 0$. Adding the elements of $O \setminus P$ one by one, diminishing returns gives

$$F(O \cup P) - F(P) \le \sum_{e \in O \setminus P} F(e \mid P) \le |O \setminus P| d \le kd. \tag{5}$$

The middle inequality follows because $d$ is the maximum marginal at $P$. The last inequality uses $d \ge 0$ and $|O \setminus P| \le k$. Since the gains are nonincreasing,

$$kd \le \sum_{i=1}^{k} a_i = F(P) - G. \tag{6}$$

Combining (4), (5) and (6) yields

$$F(O) \le F(P) + G + kd \le 2F(P).$$

The initial value $G$ in (6) cancels exactly the union-loss budget in (4). The argument allows $d = 0$ and arbitrary intersection between $O$ and $P$.

*Case 2:* $d < 0$. Every remaining gain is at most $d < 0$, so

$$H = F(E) = F(P) + \sum_{i=k+1}^{n} a_i \le F(P).$$

Using (3) and the chosen orientation $H \ge G$ gives

$$F(O) \le H + G \le 2H \le 2F(P).$$

The set $O$ was arbitrary. Thus this same order is 2-competitive at every internal cardinality. At $k = 0, n$ the unique feasible set is itself optimal, so the guarantee also holds there. Nonnegativity handles even zero endpoint values, and the proof never divides by a function value. There is no internal cardinality to consider for $n \le 1$.

*Reversing the orientation.* If $H < G$, swap $h$ and $g$ and let $(u_1, \dots, u_n)$ be the greedy order for $F'$. Its totals satisfy $H' = G \ge G' = H$, so the argument just proved applies. Write $Q_j = \{u_1, \dots, u_j\}$. A size-$k$ prefix $S_k$ of the returned reverse order is $E \setminus Q_{n-k}$. Therefore

$$F(S_k) = F'(Q_{n-k}), \qquad \mathrm{OPT}_k = \max_{|A| = n-k} F'(A).$$

The second equality uses the bijection $A = E \setminus O$ between all size-$k$ comparators and size-$(n-k)$ comparators. The swapped guarantee now gives $2F(S_k) \ge \mathrm{OPT}_k$ simultaneously for every $k$. This completes the normalized upper-bound proof.

*Nonnegative baselines.* For unnormalized nonnegative monotone inputs set $\bar{h}(S) = h(S) - h(\varnothing)$, $\bar{g}(S) = g(S) - g(\varnothing)$, and $B = h(\varnothing) + g(\varnothing) \geq 0$. The barred functions satisfy the normalized hypotheses. Apply the algorithm, including its endpoint comparison, to these functions. With $\bar{F} = \bar{h} + \bar{g} \circ (E \setminus \cdot)$, we have $F = \bar{F} + B$ and $\mathrm{OPT}_k = \overline{\mathrm{OPT}}_k + B$. Hence

$$2F(S_k) = 2\bar{F}(S_k) + 2B \geq \overline{\mathrm{OPT}}_k + 2B \geq \mathrm{OPT}_k .$$

This costs two extra baseline queries. It is the usual normalization transfer, also used in [DKLS26, Eq. (5)]; negative baselines are outside the model.

## 4 The matching lower bound from prior work

For completeness we reproduce the curvature-one coverage construction of Disser, Klimm, Lutz and Strubberg [DKLS26, Proposition 11]. For $n \geq 2$ let $E = \{a, b_1, \ldots, b_{n-1}\}$ and define

$$q(S) = \begin{cases} n-1, & a \in S, \\ |S|, & a \notin S. \end{cases}$$

This is a normalized nonnegative monotone coverage function: there are $n-1$ unit-weight features, $a$ covers all of them, and $b_i$ covers only feature $i$. Set $h = g = q$. Each summand is submodular.

At cardinalities 1 and $n-1$ the optimum is $2(n-1)$, attained by $\{a\}$ and $E \setminus \{a\}$, respectively. This value is also an upper bound because each summand is at most $n-1$. In any permutation, if $a$ is not first, its first prefix is a singleton $b_i$ and has objective value $1 + (n-1) = n$. If $a$ is first, its prefix of size $n-1$ contains $a$, and the complement is one $b_i$; again its objective value is $n$. Every permutation therefore has competitive ratio at least

$$\frac{2(n-1)}{n} = 2 - \frac{2}{n}.$$

For every $r < 2$, choose an integer $n > 2/(2-r)$. This instance has no $r$-competitive permutation, even with unlimited computation and full knowledge of both functions.

Formally, let $C_{\mathrm{struct}}$ be the infimum of uniform competitive constants of deterministic algorithms without computational restrictions, and let $C_{\mathrm{poly}}$ be the same infimum restricted to the exact polynomial-oracle model. The lower family and the upper algorithm give

$$2 \leq C_{\mathrm{struct}} \leq C_{\mathrm{poly}} \leq 2.$$

Thus both equal 2, completing Theorem 1.1. No assumption that a finite lower-bound instance attains the limiting value is needed. Normalized instances already force this lower bound, so allowing nonnegative unnormalized baselines does not alter either constant.

## 5 Scope and research methodology

The guarantee is deterministic and applies to one permutation for every cardinality, arbitrary unequal totals, and arbitrary exact ties resolved by a fixed rule. Its proof uses the pair decomposition in Lemma 3.1; it does not give a uniform guarantee for all nonmonotone submodular functions. Indeed, [DKLS26, Proposition 18] shows an unbounded deterministic incremental ratio in that larger class. The result also makes no optimality claim for narrower function classes, randomized algorithms, or additional feasibility constraints.

Substantial AI assistance was used in developing the algorithm and proof, searching for counterexamples, implementing finite exact-arithmetic and numerical diagnostic checks, comparing primary literature, and drafting this text. Counterexamples to intermediate proof proposals were used to redirect the analysis; they are not disproofs of the original ordering problem. The universal result rests on the inequalities in

Section 3, not on finite experiments or model-generated review. The proof is elementary, and no broader consequences beyond the stated incremental–decremental model are claimed.

## References

[BBB+26] Marcin Bienkowski, Joakim Blikstad, Jarosław Byrka, Martín Costa, Yann Disser, and Annette Lutz. Incremental submodular maximization: Better than greedy. In *34th Annual European Symposium on Algorithms (ESA 2026)*, volume 388 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 134:1–134:23. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2026. URL: https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.ESA.2026.134, doi:10.4230/LIPIcs.ESA.2026.134.

[CK26] Yixin Chen and Alan Kuhnle. Curvature beyond positivity: Greedy guarantees for arbitrary submodular functions. arXiv:2605.07902v1, 2026. URL: https://arxiv.org/abs/2605.07902v1, doi:10.48550/arXiv.2605.07902.

[DKLS26] Yann Disser, Max Klimm, Annette Lutz, and Lea Strubberg. Incremental–decremental maximization. *Acta Informatica*, 63, 2026. Article 31. URL: https://link.springer.com/article/10.1007/s00236-026-00546-x, doi:10.1007/s00236-026-00546-x.

## AI methodology statement

OpenAI Codex was used substantively for algorithm and proof discovery, counterexample search, computational implementation and checks, primary-literature comparison, an additional model-based examination of the argument, and preparation of the entire manuscript. These uses materially affected all sections. The model-based examination is not external peer review, and no independent human verification is claimed in this document. The mathematical claims and references require the sole human author's responsibility before any submission. Finite checks and AI-generated assessments are not mathematical certificates.